Abhishek Ravikumar

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Summary

Data professional with a Master's in Data Science and Analytics Engineering, focused on building reliable data pipelines, analytics and LLM powered systems. Hands-on with Python, SQL, PostgreSQL and Spark, and with turning raw data into dashboards (Tableau, Power BI), APIs and cloud deployed services using AWS and CI/CD. Seeking Data Analyst and Data Engineer roles.

Education

Arizona State University Tempe, Arizona
Master of Science in Data Science and Analytics Engineering; GPA: 4.0/4.0 *May 2026*

- **Coursework:** Machine Learning, Statistical Modeling, Data Mining, Big Data Analytics, Data Engineering, Database Management Systems, Distributed Computing, Deep Learning, Data Visualization, Optimization Methods

Reva University Bengaluru, Karnataka, India
Bachelor of Technology in Computer Science and Engineering; GPA: 3.6/4.0 *June 2024*

- **Coursework:** Data Structures and Algorithms, Object Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Software Engineering, Web Development, Cloud Computing

Experience

Graduate Research Assistant August 2026 – Present
Arizona State University × Honeywell *Tempe, Arizona*

- Integrated **OpenAI** and **Anthropic Claude** APIs into a **LangChain**-orchestrated Python pipeline to parse aviation safety notices, achieving **99% recall** across **10,000+** records.
- Built and maintained **FastAPI** REST endpoints with versioned schemas and WebSocket support; reviewed AI generated code (Copilot, Cursor) for security and correctness before merging.
- Tuned **PostgreSQL** queries and indexing, cutting p95 latency by **20%**.
- Automated CI/CD via **GitHub Actions** on **AWS**, enabling **100+** peer code reviews with zero regressions across **6+** sprint releases.

Data Analyst & Full Stack Developer Intern June 2025 – July 2026
Datasoothra *Tempe, Arizona*

- Collaborated with a cross functional team to build a responsive client business website using **React, HTML and CSS**, translating design requirements into mobile friendly UI components.
- Built multi page navigation, a contact form, testimonials section and image gallery, working with **Node.js** backend developers to connect form submissions.
- Performed exploratory data analysis on large structured datasets with **Python (Pandas, NumPy)** to uncover trends and business insights.
- Designed and automated reporting dashboards with **Excel** and **Python**, improving reporting efficiency by **30%**.
- Wrote **SQL** queries (JOINs, subqueries, aggregations) to extract, transform and analyze relational data for business reporting.

Data Analyst Intern October 2022 – May 2024
MEVI Technologies LLP *Bengaluru, India*

- Engineered and validated structured datasets from **20,000+** website records using **SQL** (joins, window functions) and **Python (Pandas)**, with deduplication, null handling and feature engineering for phishing risk analysis.
- Conducted EDA with **Matplotlib** and **Seaborn** to find high risk URL and domain patterns, improving phishing detection accuracy by **12%**.
- Designed interactive **Power BI** dashboards with drill through filters and KPI tracking, presenting weekly risk trends to IT security and management.

Projects

RideAI | *Java, Spring Boot, Next.js, Kafka, Redis, Docker, GitHub Actions, Testcontainers, Sentry*

- Built a real time ride hailing platform end to end, with a **Java, Spring Boot** backend and a **Next.js** frontend.
- Used **Redis GEO** for driver matching, sustaining about **4,900 GPS updates/s** at **p95 34 ms** with **0% errors** across 100 simulated drivers.
- Designed event streaming with **Kafka** to move ride and location events between services.
- Set up CI/CD with **GitHub Actions**, integration tests with **Testcontainers**, containerization with **Docker**, and monitoring with **Sentry**.

E-Commerce Analytics Pipeline with GenAI Insights Assistant | *Python, PostgreSQL, SQL, Tableau, Chroma, Claude API*

- Designed a **PostgreSQL** relational schema from raw multi table e-commerce data, with **SQL** analyses and **Tableau** dashboards on delivery performance, revenue by category and regional trends.
- Added a **RAG** layer for natural-language querying, embedding precomputed metric summaries with **Chroma** and a local sentence embedding model.
- Built a retrieval augmented QA loop with the **Claude API**, tuning retrieval depth to eliminate hallucinated answers across a curated test set.
- Published the full pipeline with documentation on GitHub.

A Data-Driven Approach for Detecting AI-Generated Images | *Python, PyTorch, ResNet-50, EfficientNet-B3, FastAPI, React, Grad-CAM*

- Built an AI image forensics system with **ResNet-50, EfficientNet-B3** and custom CNNs: **91.4% accuracy** and **0.968 AUC** on ˜28K real and AI-generated images.
- Designed a wavelet based frequency domain CNN; ablation studies improved performance **4.86%** over a pixel space baseline.
- Ran robustness and explainability analysis with Grad-CAM, ROC-AUC, test-time augmentation and image degradation testing.
- Deployed a **React and FastAPI** app with ensemble inference, Grad-CAM visualizations and forensic reporting, using the **Claude API** for natural language report summaries.

Neighborhood Risk & Change Analytics Pipeline | *R, Mclust, Shiny, Regression*

- Built a geospatial **dorling cartogram** and interactive **Shiny** dashboard over a metro area's census tracts, deployed on **shinyapps.io**.
- Applied **Mclust** model based clustering to segment neighborhoods into distinct groups.
- Built predictive and explanatory **regression** models of median home value change from 2000 to 2010.
- Ran diagnostics for multicollinearity, skew and effect size to validate the models.

Morse Code-Based Secure Authentication System using Eye Blink Recognition | *Python, OpenCV, Haar Cascade, LBPH*

- Built a computer vision system that translates eye blink patterns into Morse code for secure user verification.
- Implemented face and eye detection with **OpenCV Haar Cascades** and **LBPH** for real-time recognition.
- Engineered blink pattern detection logic to reduce false authentication attempts.
- Recognized and funded by the **Karnataka State Council for Science and Technology (KSCST)**.

Letter – Minimalist Programming Language | *Python*

- In a **4-person team**, built an interpreted language from scratch with single uppercase letter keywords and a hand written lexer, parser and AST evaluator.
- Built the lexer, handling typed variables, operators and literals.
- Wrote **10+** test files covering type mismatches, undefined variables and out of bounds access across loops, functions and data structures.
- Contributed if-else, for/while loops, functions with parameters and return values, constants and a ternary operator.

Technical Skills

Languages: Python, SQL, JavaScript, Java, C, R
Generative AI: LLMs, Prompt Engineering, RAG, RLHF, LLM Fine-Tuning, Agentic AI (CrewAI), Streamlit, LangChain, Chroma
ML & Data Science: Deep Learning, Statistical Modeling, Data Mining, Clustering, Regression, Model Evaluation, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, OpenCV
Backend & Web: FastAPI, Flask, Spring Boot, React, Next.js, Node.js, Kafka, Redis
Databases & Big Data: PostgreSQL, MySQL, MongoDB, Apache Spark, Hadoop, Databricks
Visualization: Tableau, Power BI, Shiny, Matplotlib, Seaborn, Plotly
Tools & DevOps: Git/GitHub, GitHub Actions, Docker, Testcontainers, Sentry, AWS, Jupyter

Certifications

- AWS Certified Cloud Practitioner
- Generative AI with Large Language Models (DeepLearning.AI)
- Agentic AI (DeepLearning.AI)
- Fast Prototyping of GenAI Apps with Streamlit (DeepLearning.AI)
- Python for Data Science, AI & Development (IBM, Coursera)
- Programming for Everybody, Getting Started with Python (University of Michigan, Coursera)